Evidence Acquisition and Analysis: Hard Drives

Introduction

In this practical you will perform hard drive data acquisition by using AccessData FTK imager and a hardware write blocker.

By the end of this lab you will be able to

- Use a hardware write blocker, ensuring the integrity of suspect's digital data is protected
- Use AccessData FTK imager to acquire to images in a forensic sound manner
- Describe the standard hard drive acquisition process
- Perform basic analysis on forensic images within AccessData Imager

Task 1: Data acquisition

plan : use FTK Imager to create an image for a folder.

This is a group exercise and each group has 4 persons. Ask the lecturer if you cannot find a group.

- Find a hard drive on the workbench in the middle of the lab
- Get a write blocker with all associated cables from the lecturer
- Logon to Windows OS and locate the AccessData FTK Imager application by clicking the Windows start button - > All Programs - > AccessData - > FTK Imager

Follow the instructions to create a forensic image

1. Connect the write- blocker to the main (the "DC in" light should be on)
2. Carefully connect the hard drive with the write- blocker via a suitable interface and then connect the write- blocker to the forensic workstation via the USB connection
3. Switch on the write- blocker and make sure the "write block" light is on (what if the light is not on?)
4. Launch the AccessData FTK imager application
5. Go to File --> Create Disc Image,
6. In the popup window, choose the "Physical Drive" option and click on "Next"

- Have an explore on other options if you have time

7. Select the correct hard drive that need to be imaged and click “Finish”
8. In the new popup window, click on “Add” and Select Raw(dd) for the Destination Image Type
    - How about other types (e.g. SMART, E01 and AFF), what do they do?
9. For the Evidence Item Information, complete as much and accurate as you can; and then click on “Next”
10. Choose a Destination Folder (better to put on Desktop) and give a meaningful image filename
11. Regarding the “image fragment size”, “compression”, and “use AD Encryption” options, use them based upon your own interpretation and Click “Finish”
12. Click finish
13. Once the image is created, note down the sector count, MD5, and SHA1 hash values.

Task 2: Data analysis

This is an individual exercise; and you will use AccessData FTK imager to analyse the image that you just created by using these steps:

1. Go to “file” --> “Add Evidence Item”
2. Choose “image file”, locate the image file and click “Finish”

Explore the image in the evidence tree pane and answer the following questions (hint: check the Properties pane)

1. How many bytes per sector for the image?

2. How many sectors in total on the image?

3. What is the type of the image?

4. How many partitions does the image have?

5. What is the starting sector number of the NTFS partition?

6. Within the NTFS partition, in the root folder, locate the file "$MFT", what is the MFT record number for this file? (hint: check the NTFS information within the Properties pane)

7. How many Windows users can you find and what are their names?

8. Locate the readme.txt file in the "Documents" folder of the user Adam. What is the Resident attribute value of this file? (hint: check the NTFS information within the Properties pane)

9. Within the same folder, which file is encrypted and what kind of encryption method is used?

10. Within the same folder, which files do you want to investigate more? And why? What are the contents of these files (hint: you may want to export these files to your desktop to perform further analysis)

Task 3: Further analysis

11. How many files can you find in the $Recycle.bin folder and what are they?

12. Which web browsers does Adam use and can you list the websites he visited?

13. What kind of files can you find in Adam's Downloads folder? What do they suggest and why?

14. Identify 4 applications that are installed on this image?